

Teddy Bear Wheelchair Individual Final Report

Section 07 and Team 8

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Statement:

This report was completed by Deep Patell alone, in collaboration with my group members, as part of my TBWC project requirements.

Name: ____Deep Patel____

Signature: ________

1. Introduction

Throughout the TBWC project, my role expanded far beyond my originally assigned responsibility of design and aesthetics. While I began as the co-lead for the aesthetic subsystem alongside Humza, I quickly became involved in every major component of the prototype, ultimately becoming the primary contributor to the design, construction, troubleshooting, and integration of the entire system.

A major portion of my responsibility became the physical build and structural development of the wheelchair. Although the chassis was assigned to Ryan and Brendan, both were frequently unavailable or disengaged during work sessions. To prevent delays and ensure our team met deadlines, I took full ownership of the structural build. This included measuring, assembling, mounting, and adjusting the Meccano frame, as well as creating stable supports for the wheels, motor, and launcher. I completed the entire chassis assembly myself, with occasional assistance from Sami when extra hands were needed.

In addition to the chassis, I fully designed and constructed the entire aesthetic ambulance shell, ensuring that the body fit securely over the wiring and electronics while providing safety clearances around the launcher and moving components. This involved multiple iterations, cutting and shaping cardboard, creating structural bracing, and ensuring that the teddy bear seating area was properly supported. The final aesthetic design required careful alignment with the mechanical build, and I completed all of this independently due to limited participation from Humza, who only attended one full meeting during the build phase.

I was also heavily involved in testing, troubleshooting, and performance refinement. I conducted numerous forward/backward speed tests, launcher tests, wheel alignment checks, and clearance trials to identify inconsistencies. I took the TBWC prototype home multiple times to make progress outside scheduled meetings, ensuring the design was always improving between team work sessions. This additional independent work was crucial in keeping the project on track.

Beyond technical tasks, I assumed the central leadership and coordination role for our team. I organized every meeting, ensured scheduling worked for as many members as possible, and communicated expectations and weekly goals. I consistently reminded the group of deadlines, clarified tasks, answered questions, and redistributed work when necessary. My leadership role grew because the workload across members became imbalanced. Some teammates, such as Ayaan, consistently completed their electrical and coding responsibilities, but others struggled to maintain engagement. To avoid falling behind, I stepped in repeatedly to cover uncompleted tasks and integrate all subsystems into a functioning prototype.

The intention was for each pair to independently develop their subsystem and bring it together during integration sessions. Early on, I helped structure this division so that responsibilities were clear and evenly distributed.

For meetings, I took the initiative to establish a consistent, reliable schedule, organizing sessions every Tuesday after 5:00 PM and Thursday after 6:00 PM, with extra sessions added when work was behind or deadlines were near. I communicated meeting reminders, confirmed attendance, and adjusted times whenever teammates had schedule conflicts. During meetings, I acted as the facilitator, checking on each subsystem's progress, ensuring tasks were being completed, and helping teammates whenever they encountered challenges.

However, as the project progressed, our planned division of labour did not hold. Team engagement varied significantly. Ayaan maintained full accountability for the electrical and coding work, and Sami contributed when he was present, during mechanical assembly. But participation from others was inconsistent, some members rarely attended meetings, struggled to begin their assigned tasks, or needed repeated guidance. To prevent delays, I took the responsibility of completing unassigned or incomplete portions of the build.

Because of this shift, our team's work approach evolved into a hybrid structure, where I coordinated tasks based on urgency rather than the initial contract. I handled the mechanical and aesthetic responsibilities almost entirely, while ensuring electrical and software components from Ayaan were correctly integrated into the chassis. My leadership ensured that despite inconsistent effort distribution, the team remained aligned enough to build a fully functional prototype by the deadline.

As outlined in our original team contract, we had planned to divide the TBWC workload based on each member's strengths. Humza and I were assigned to the aesthetics, focusing on the visual design, theme, and overall appearance of the wheelchair. Ryan and Brendan were responsible for the chassis, including measurements, structural stability, and building the base frame. Sami and Ayyan were designated to work on the electrical components, such as wiring, Arduino integration, and the servo mechanisms. Although these task divisions did not fully play out as intended, they represented our initial strategy for organizing responsibilities and ensuring that every part of the project had a dedicated focus.

Team Member Contribution

Ayaan Ahsan — Electronics Lead

Ayaan consistently fulfilled his responsibilities as the Electronics lead, completing all required wiring, coding, and Arduino setup. He took initiative in troubleshooting electrical issues and provided reliable support whenever the chassis or launcher needed integration with the electrical subsystem. His dependable contributions ensured that the motor system and control logic functioned correctly throughout the project.

Sami Cherniak — Electronics Support

Sami participated in electrical work when present and assisted with wiring and occasional chassis assembly tasks. While his involvement was inconsistent and he required direction to remain on task, he provided useful help during hands-on building sessions, particularly when additional support was needed. His contributions, though limited, were constructive when he was engaged.

Brendan Hunter — Organization Manager

Although Brendan was assigned a leadership and organization role, his participation did not match the expectations outlined in our team contract. He rarely initiated tasks independently and contributed minimally to the chassis and launcher subsystems he was responsible for. While he attended some sessions, he required constant redirection, and his limited engagement placed additional workload on the rest of the group.

Ryan Grant — Construction Lead

Ryan's attendance and participation were significantly below what was agreed upon in the team contract. As the designated Construction Lead, he contributed very little to the chassis development, which required other teammates to assume full responsibility for the build. His frequent absence and lack of follow-through directly impacted workload distribution and placed pressure on others to complete the subsystem.

Humza — Aesthetics Support

Humza was assigned as co-lead for the aesthetics subsystem but attended only one major work session and contributed minimally to the design and build process. While he offered some initial ideas, he did not participate in the construction or refinement of the aesthetic shell, which required the remaining work to be completed independently. His limited availability resulted in an unequal division of work and required others to take over his responsibilities.

Self-Evaluation — Deep Patel

I consistently exceeded my assigned responsibilities by completing the full construction of the chassis, launcher, and aesthetic systems, as well as managing the majority of testing, troubleshooting, and subsystem integration. I also acted as the team coordinator by organizing meetings, setting deadlines, and maintaining communication to keep the project progressing. My leadership and independent work were essential in ensuring the prototype was completed on time and met all functional requirements.

2. Contribution Elements

Project Aspect	Design Project Phase and Contribution.
<i>Subsystem 1 – Rolling Chassis</i>	Empathize with People I ensured the rolling chassis was designed with stability and comfort for the teddy bear in mind by creating a firm, level base that supported the seat and kept wiring safely below. I made sure the chassis geometry prevented vibrations or sudden jerks so the user experience would be safe and smooth. (Logbook: chassis fit checks, seat clearance notes Entry 8-9) Define the Problem I clearly defined the key constraints for the rolling chassis: structural stability, safety around wiring, minimal mass, and compatibility with the motor and pulleys. I also helped define the need for a symmetric, rigid frame to avoid drift and maintain a straight path during the speed run. (Logbook: early design notes, mass calculations Entry 12-13) Generate Ideas I developed the main structural layout for the chassis, proposing the frame width, wheel placement, motor mounting location, and pulley alignment ideas before building them myself. I also ran these ideas through the group and incorporated feedback when possible, even when participation was low. (Logbook: brainstorming sketches, chassis concept discussion Entry 6) Prototype I was fully responsible for physically building the rolling chassis, assembling the Meccano frame, mounting the wheels and axles, positioning the motor, and aligning the drive train. I refined the structure multiple times as issues arose, improving rigidity and adjusting fit to support later subsystems. (Logbook: build entries over multiple sessions Entry 10-11, 12-15) Testing Solutions I repeatedly conducted test runs with Ayaan to assess chassis movement, drift, torque balance, and wheel traction. Through these tests, I identified that the pulley offset caused drift and that the chassis needed reinforcement to prevent frame twist under load. I also tested acceleration, braking, and speed timing to verify performance. (Logbook: test notes, timing results Entry 24) Troubleshooting & Iteration

	I troubleshoot issues with drifting, pulley alignment, axle straightness, and motor mounting, adjusting the chassis until the motion was consistent. I also helped diagnose Arduino timing problems and integrated code changes with mechanical adjustments to ensure predictable movement. These iterations continued across several meetings and at home. (Logbook: troubleshooting notes, timing adjustments Entry 24) Assess Results I evaluated the chassis performance after each test, identifying what improved and what still needed adjustment. Based on the results, I recommended adding time to the speed run and realigning the chassis before final testing. I interpreted test outcomes to ensure the final system met stability and speed requirements. (Logbook: reflections on test-day performance Entry 23-24)
<i>Subsystem 2 – Launching Mechanism</i>	Empathize with People I ensured that the launching mechanism was designed in a way that did not compromise the teddy bear's safety by positioning it on the side of the chassis, far from the seating area and wiring. I also designed the launcher so that the arm's motion path stayed clear of all passengers and components throughout firing. (Logbook: launcher Entry 10-11) Define the Problem I identified the key problem objectives for the launcher: achieve consistent range, maximize energy output, maintain safety clearances, and integrate the mechanism without interfering with wheels or wiring. I also defined functional constraints such as launcher height, arm length, mounting stability, and trigger reliability. Generate Ideas I personally generated all initial launcher concepts, experimenting with mousetrap-only builds, elastic-assisted designs, and hybrid tension systems. I evaluated multiple arm lengths, mounting angles, and release methods before settling on the final approach. I ran each idea through the team and incorporated feedback whenever possible, despite limited participation. (Logbook: launcher concept sketches, elastic test notes Entry 6-7) Prototype I completely built the launching mechanism myself, including the mousetrap mounting, arm attachment, hooking bar design, elastic tension system, and integration into the chassis frame. I created the

	projectile funnel using cut plastic and ensured the mechanism was structurally supported on the side panel. (Logbook: build, launcher assembly entries, Entry 10-12) Testing Solutions I performed extensive launcher testing both in lab/meetings and at home, measuring projectile distance, height, and consistency across dozens of trials. I tested variations in elastic tension, arm angle, hooking bar position, and projectile mass. These tests allowed me to determine the best configuration for range and reliability. (Logbook: launcher test results, range measurements Entry 22-24) Troubleshooting & Iteration I troubleshot repeated failure modes such as the hooking bar catching on the servo, insufficient arm retraction due to elastic tension, upward bending of the arm under load, and inconsistent projectile release. Through iterative rebuilding, repositioning the servo, adding reinforcement, adjusting tension, and modifying the funnel, I significantly improved launch accuracy. (Logbook: troubleshooting notes, servo reposition entries (Entry 24)) Assess Results After each series of test launches, I assessed the accuracy, power output, and stability of the launcher, identifying the root causes of inconsistencies. I used these results to finalize elastic selection, arm adjustment, and servo placement, ensuring the system performed safely and predictably during final testing. (Logbook: launch performance analysis Entry 23-24)
<i>Aesthetics & Accessibility</i>	Contribution Summary I was fully responsible for the design, construction, and completion of the aesthetics subsystem. Even though this subsystem was originally assigned to both Humza and me, I completed the entire aesthetic build myself and took the prototype home multiple times to finish all required work. I designed the box-shaped ambulance shell, constructed it from cardboard and cardstock, added all colour layers and detailing, and ensured it integrated safely with the chassis and launcher. Idea Generation I developed all aesthetic concepts for the final design, transitioning from early “fun” ideas (Batmobile, Minecraft cake, Krabby Patty) to a fully functional ambulance theme. I chose a box-like exterior shape for structural strength, ease of construction, and compatibility with the

	chassis footprint. I also created the screw-mounting idea to attach the exterior securely without adhesives, which solved a key constraint of the project. (Logbook: aesthetic concept sketches and notes Entry 6-7, 10) Prototype / Build I independently built the entire aesthetic shell using cardboard, construction paper, and cardstock, ensuring precise dimensions so that the shell fit cleanly over the wiring and electrical components while maintaining accessibility for repairs. I cut, shaped, reinforced, and assembled all panels by hand, and I screwed the frame sections together to meet project restrictions and ensure long-term durability. (Logbook: aesthetic construction entries, photos) Accessibility Considerations In designing the shell, I ensured that the teddy bear passenger had full visibility, proper chest-level support, and a stable seated position. I preserved clearance above the wiring and left space around the launcher to prevent interference during firing. I also kept the top section removable so we could access the Arduino, breadboard, and wiring during testing. (Logbook: clearance and seat-support measurements Entry 15, 21-22) Testing & Adjustments After building the shell, I tested it repeatedly on the chassis to verify fit, stability, and weight impact. I adjusted panel widths, reinforced weak areas, and ensured the aesthetic layer did not introduce vibrations or restrict movement. I evaluated accessibility during wiring changes and made modifications to keep all electrical components reachable. (Logbook: shell fit checks and modification notes Entry 21-23) Final Assessment I assessed the completed aesthetic subsystem to confirm it met all design requirements: safety, accessibility, structural integrity, and thematic cohesion. The final ambulance design was visually clear, durable, and fully compatible with the mechanical and electrical subsystems, ensuring the prototype remained functional and easy to service.
<i>Team Contract Document</i>	Although our contract outlined clear expectations, such as equal division of responsibilities, consistent attendance at twice-weekly meetings, and timely completion of subsystem tasks, these expectations were not upheld by most team members. As a result, I had to assume the responsibilities originally assigned to others, including building the chassis, launcher, and full aesthetics, while also managing scheduling,

	communication, and documentation as stated in the contract. Only Ayaan consistently fulfilled his assigned electronics role, so I effectively acted as the organizer, construction lead, and aesthetics manager simultaneously to ensure our team met deadlines and delivered a functional prototype.
<i>Additional Contributions</i>	Project Management I took full responsibility for managing the project timeline, scheduling meetings, tracking progress, and ensuring tasks were completed before deadlines. I consistently communicated expectations, clarified responsibilities, and adjusted plans when work from others fell behind. My oversight kept the project on schedule and prevented delays that could have jeopardized the final prototype. Team Coordination I acted as the central coordinator by answering team questions, guiding workflow during meetings, and distributing tasks when others were unsure of their roles. I ensured all subsystems stayed aligned by regularly checking in with each member and integrating mechanical, electrical, and aesthetic components into a cohesive design. This coordination held the group together and kept the project moving forward. Conflict & Accountability Management When team engagement was inconsistent, I maintained professionalism by redirecting focus, providing reminders, and reinforcing the expectations outlined in our contract. I addressed issues calmly and constructively, ensuring the project environment remained productive despite uneven contributions. I held teammates accountable while stepping in to complete essential tasks when necessary to ensure project success. Technical Leadership In addition to my assigned role, I provided hands-on technical support across all subsystems, helping with electrical troubleshooting, mechanical adjustments, coding discussions, and launcher alignment. I independently completed major structural and aesthetic work when others were unable to do so. My technical leadership ensured every subsystem was functional, safe, and integrated properly. Independent Work Outside Meetings To keep the project on track, I repeatedly took the TBWC home and continued significant construction, testing, and refinement outside scheduled team sessions. This additional effort allowed me to solve

	mechanical issues, test launcher performance, and complete the full aesthetic shell without delaying group progress. These independent contributions were critical to achieving a polished and functional final prototype. (Logbook Entry 10,11,21,24)
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